A Dissertation-Style Research Presentation

Machine Learning and Generative AI Systems for Cross-Asset Liquid Financing

Five Production-Grade Case Studies: Securities-Lending Fee Forecasting, Client Margin Optimization, Funding-Spread Anomaly Detection,

Deep-Learning Balance Forecasting, and a Retrieval-Augmented Financing Copilot

Candidate Presentation

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Take-Home Discussion

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Abstract

This work presents five production-oriented machine learning and generative AI systems designed for a cross-asset Liquid Financing platform spanning Equities & Delta One, Rate and Credit Financing, FX, Risk, and Futures & Prime Derivatives Solutions. Each system is scoped directly against a stated business roadmap of 50+ identified problem statements and mirrors the explicit technical requirements of the target role: regularized regression (OLS, LASSO, Elastic Net), tree-ensemble methods, time-series forecasting, deep sequence models (RNN/LSTM/GRU), and generative AI (fine-tuning, prompt engineering, retrieval-augmented generation, and evaluation). We present, for each system: (i) the business problem and P&L or risk lever it addresses; (ii) the mathematical formulation and model architecture; (iii) a reference implementation; and (iv) a validation and production-monitoring protocol consistent with institutional model-risk practice. We conclude with a cross-cutting discussion of deployment standards, model governance, and the trade-offs between prompt-engineered and fine-tuned generative systems in a regulated financial environment.

Contents

Chapter 1

Introduction and Problem Framing

1.1 Business Context

Liquid Financing intermediates the price of *borrowing* — securities, cash, and balance sheet — across a cross-asset platform. Its economics are driven by three coupled quantities: (1) the fee or rate charged for financing an asset, (2) the collateral haircut required to protect against counterparty default, and (3) the balance-sheet capacity consumed by client positions. Each of the five case studies in this dissertation targets exactly one of these economic levers, chosen so that, collectively, they cover every technical bullet point in the target job description while remaining traceable to a concrete desk P&L or risk outcome.

1.2 Mapping to the Stated Technical Requirements

Case Study	**Core Method**	**JD Requirement Addressed**
P1	Elastic Net + Gradient-Boosted Trees + AR/event time-series blend	Regression (OLS/LASSO/Elastic Net), Tree models, Time series
P2	LASSO haircut model + GBM correlation-breakdown add-on	Regression, Tree models, model-risk governance
P3	Robust Mahalanobis + Isolation Forest + LLM text fusion	Time series, Gen AI
P4	Sequence-to-sequence GRU/LSTM with quantile heads	Deep Learning (Neural Networks, RNN, LSTM, GRU)
P5	Hybrid retrieval-augmented LLM copilot	Gen AI: fine-tuning, prompting, RAG, evaluation, inference infrastructure

Chapter 2

P1 — Securities-Lending Fee and Rebate-Rate Forecasting

2.1 Problem Statement

We forecast the daily specialness fee $f_{i,t}$, expressed in annualized basis points over the general collateral (GC) rate, for a universe of hard-to-borrow equities, at horizons $h \in \{1, \ldots, 5\}$ business days, to feed a pre-trade financing pricing engine.

2.2 Model Formulation

A three-tier ensemble is used. The first tier is an Elastic Net regression:

$$\hat{\beta} = \arg\min_{\beta} \|y - X\beta\|_2^2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2 \tag{2.1}$$

chosen for auditability: every surviving coefficient can be defended in a model-risk review. The second tier is a gradient-boosted tree ensemble,

$$F_M(x) = \sum_{m=1}^{M} \gamma_m h_m(x), \qquad h_m = \arg\min_h \sum_i L\big(y_i,\, F_{m-1}(x_i) + h(x_i)\big), \tag{2.2}$$

which captures the empirically convex relationship between utilization and fee once utilization exceeds approximately 90%, a threshold effect that a purely linear model cannot represent. The third tier is a mean-reverting, event-augmented time-series component,

$$f_{i,t} = \mu_i + \phi\,(f_{i,t-1} - \mu_i) + \sum_j \kappa_j \nVdash[\mathrm{event}_{j,t}]\, e^{-\eta_j (t - t_j)} + \varepsilon_t, \tag{2.3}$$

which explicitly models the predictable fee spikes around dividend record dates and index rebalancing events. The final forecast is a constrained convex combination of the three tiers, with weights fit by non-negative least squares on a walk-forward validation window.

2.3 Reference Implementation (excerpt)

```
en = ElasticNetCV(l1_ratio=[.1,.5,.7,.9,.95,1], cv=5).fit(X_train, y_train)
gbm = LGBMRegressor(n_estimators=600, num_leaves=31,
                    learning_rate=0.03).fit(X_train, y_train)
w, _ = nnls(np.column_stack([en.predict(X_val), gbm.predict(X_val)]), y_val)
forecast = np.column_stack([en.predict(X_new), gbm.predict(X_new)]) @ (w / w.sum())
```

2.4 Validation Protocol

Weighted MAPE (notional-weighted), directional hit-rate on fee changes exceeding 25 bps, and pinball loss at the 10th/90th percentiles are tracked on an expanding walk-forward window. Feature population stability index (PSI) is monitored monthly, with automatic fallback to the Elastic Net tier if the gradient-boosted tree's input PSI exceeds 0.25.

Chapter 3

P2 — Client Margin and Haircut Optimization

3.1 Problem Statement

Given a client's cross-asset collateral basket, we predict the required per-asset haircut and portfolio-level initial margin such that the modeled 99% five-day liquidation shortfall is covered, while minimizing over-collateralization relative to the existing rules-based haircut grid.

3.2 Model Formulation

The per-asset haircut is modeled log-linearly with LASSO-selected covariates,

$$\log h_i = \beta_0 + \beta_1 \log \sigma_i + \beta_2 \log(\mathrm{ADV}_i) + \beta_3 \,\mathrm{Corr}_{i,\mathrm{mkt}} + \beta_4 \,\mathrm{Rating}_i + \varepsilon_i, \tag{3.1}$$

and the portfolio-level required initial margin is decomposed into a linear per-asset component plus a tree-learned correlation-breakdown add-on,

$$\mathrm{IM}_{\mathrm{portfolio}} = \sum_i h_i \,|q_i|\, P_i \;+\; g_\theta(\mathbf{q}, \boldsymbol{\Sigma}, \text{concentration}), \tag{3.2}$$

where g_θ is a gradient-boosted regressor trained on the historical-simulation shortfall residual unexplained by the linear component. This add-on is precisely the mechanism that captures correlations moving toward one under stress, historically the primary failure mode of purely rules-based haircut grids. The final output is clipped to [rules-based floor, $1.2 \times \mathrm{VaR}_{99\%}$] and trained against the chance constraint

$$\Pr\left(\mathrm{Shortfall}_{5d} > h \cdot V\right) \leq 1\%. \tag{3.3}$$

3.3 Backtesting Discipline

Kupiec proportion-of-failures and Christoffersen independence tests are applied to daily VaR breach counts. Champion/challenger comparison runs for a minimum of two full quarters in shadow mode before promotion, with stress overlays replaying 2008, March 2020, and 2023 regional-banking scenarios against the current book.

Chapter 4

P3 — Cross-Asset Funding-Spread Anomaly Detection

4.1 Problem Statement

We detect anomalous divergence across the repo, securities-lending, cross-currency-basis, and credit-basis complex in real time, fused with a natural-language classifier over overnight desk commentary, and rank alerts by likely P&L impact.

4.2 Model Formulation

A robust multivariate anomaly score is computed via Mahalanobis distance under an EWMA-updated, Minimum-Covariance-Determinant-robustified covariance estimate,

$$z_t = (\mathbf{x}_t - \boldsymbol{\mu}_t)^\top \boldsymbol{\Sigma}_t^{-1} (\mathbf{x}_t - \boldsymbol{\mu}_t), \tag{4.1}$$

cross-checked by an Isolation Forest score $s(x) = 2^{-E[h(x)]/c(n)}$ that is more robust to regime-dependent non-Gaussianity. A fine-tuned or prompt-engineered language model classifies overnight commentary into stress categories, and the statistical and textual signals are fused as

$$\mathrm{AlertScore}_t = \alpha \cdot \mathrm{sigmoid}(z_t - z^*) + (1 - \alpha) \cdot p(\mathrm{stress} \mid \mathrm{text}_t). \tag{4.2}$$

4.3 Human-in-the-Loop Design

Every alert above threshold routes to a trader queue accompanied by the driving spread chart, the top-attributed statistical features, and a cited natural-language rationale; no action is executed automatically. Precision and recall are tracked weekly against trader-labeled outcomes, with the alert threshold retuned via a Neyman–Pearson-style asymmetric cost ratio reflecting that missed funding-stress events are materially more costly than false positives on this desk.

Chapter 5

P4 — Prime Balance and Utilization Forecasting via Deep Sequence Models

5.1 Problem Statement

We forecast daily aggregate client financing balances and utilization ratios one to twenty business days ahead, with explicit treatment of month-end and quarter-end window-dressing effects, for Treasury capacity and RWA/LCR planning.

5.2 Model Formulation

A sequence-to-sequence Gated Recurrent Unit (GRU) encoder-decoder is used,

$$h_t = \mathrm{GRU}(x_t, h_{t-1}), \qquad \hat{y}_{t+1:t+H} = \mathrm{Decoder}(h_T,\ \text{calendar embeddings}), \tag{5.1}$$

with an explicit, learned calendar embedding concatenated at every decoder step so that the known, calendar-driven quarter-end balance-reduction pattern is treated as an inspectable input rather than left for the network to rediscover implicitly. Three quantile heads are trained jointly via the pinball loss,

$$\mathcal{L} = \sum_{q \in \{0.1, 0.5, 0.9\}} \sum_{h=1}^{H} \rho_q\big(y_{t+h} - \hat{y}_{t+h}^{(q)}\big), \qquad \rho_q(u) = u\,(q - \mathbb{1}[u < 0]), \tag{5.2}$$

producing a P10/P50/P90 balance forecast band rather than a false-precision point estimate. An LSTM variant is retained as a challenger given its stronger long-memory gating.

5.3 Baseline Discipline

The GRU architecture is required to outperform a seasonal-naive baseline, a SARIMAX model with an explicit quarter-end dummy, and a well-tuned LightGBM model on lagged and calendar features, on walk-forward mean absolute percentage error and pinball loss, before its added production and serving complexity is justified to the panel.

Chapter 6

P5 — A Retrieval-Augmented Generation Copilot for the Financing Desk

6.1 Problem Statement

We build a retrieval-augmented generation (RAG) system over internal financing policy documents, desk commentary, and the structured outputs of the P1–P4 models, that answers trader queries with citations and explicitly declines to answer when evidence is insufficient.

6.2 Retrieval Formulation

Hybrid sparse-dense retrieval is used,

$$\mathrm{score}(q, d) = \lambda \cdot \mathrm{BM25}(q, d) + (1 - \lambda) \cdot \cos\big(E(q), E(d)\big), \tag{6.1}$$

followed by cross-encoder reranking of the top candidates, with λ tuned against a human-labeled query/document relevance set using recall@10 as the primary retrieval metric.

6.3 Fine-Tuning versus Prompt Engineering: An Explicit Trade-off

Per the target role's requirement to *evaluate and articulate trade-offs* between modeling approaches, we adopt a staged strategy: a prompt-engineered, retrieval-augmented system is deployed first to establish an evaluation harness and collect production query logs; intent classification and domain-jargon normalization are subsequently migrated to a small, LoRA-fine-tuned model once a sufficient labeled query corpus exists, while final-answer synthesis remains on a frontier base model with retrieval for auditability. This reflects a governance-first calculus: full fine-tuning offers superior domain fluency and lower marginal cost per query, but at the expense of update latency and audit-trail complexity relative to a prompt-plus-retrieval design whose full reasoning trace is visible at inference time.

6.4 Evaluation Harness

$$\mathrm{Faithfulness} = \frac{\#\{\text{claims in the answer supported by retrieved context}\}}{\#\{\text{claims in the answer}\}}. \tag{6.2}$$

Faithfulness, retrieval recall@10, human-graded answer correctness against a quarterly-refreshed golden set, and refusal precision on out-of-scope queries are tracked continuously, with a p95 latency service-level objective of under four seconds during trading hours.

6.5 Inference Infrastructure

Consistent with the target role's requirement to configure inference infrastructure, non-public, client-identifying, or desk-proprietary queries are routed to an on-premises, BMC-hosted open-weight model behind a request-routing layer that enforces entitlement and PII filtering prior to prompt assembly; only public-market-context queries are eligible for routing to an external frontier model endpoint.

Chapter 7

Cross-Cutting Production and Governance Standards

7.1 Validation

All five systems are validated exclusively on walk-forward, expanding-window splits; random cross-validation is explicitly avoided given the autocorrelated, event-clustered structure of financing time series, which would otherwise leak information across the train/validation boundary.

7.2 Model Risk Documentation

Each system carries a standard model-risk package: statement of purpose, conceptual soundness review, developmental evidence (in-sample and out-of-sample performance), independent outcomes analysis (backtesting), and an ongoing performance-monitoring plan, with a minimum two-quarter shadow period prior to any production promotion.

7.3 Monitoring and Fallback

Feature-level population stability index, prediction drift, and backtest breach counts are monitored continuously, with automatic fallback to the prior champion model on threshold breach, and canary release to a 5% traffic slice prior to full production rollout.

Chapter 8

Conclusion

The five systems presented — fee forecasting, margin optimization, anomaly detection, deep sequence-model balance forecasting, and a retrieval-augmented financing copilot — jointly span every modeling technique enumerated in the target job description, are each traceable to a specific desk P&L or risk lever, and are each specified with a production validation and governance protocol suitable for a regulated cross-asset financing business. Collectively they represent a candidate execution plan against the stated backlog of fifty-plus prioritized problem statements, beginning with the ten already flagged as active priorities.